L-RAIL Scientific Concept and Mission

Omar Ariel Vallejos (Independent Researcher)
Lunar Radio-interferometry & Astrophysics Infrastructure Laboratory
VCV48 Framework – For CoPAG, PhysPAG

Abstract number: 128

Scientific Background and Gap

The VCV48 Crystalline Vacuum Model:

- Postulates quantum vacuum possesses discrete, anisotropic geometric structure
- Complete octahedral symmetry (O_h, order 48)
- Projection onto plane of sky reduces to hexagonal symmetry (D_6)
- Induces periodic 60° azimuthal modulation (Fourier harmonic $m = 6$)

Detection Method:

- Observable in Electric Vector Position Angle (EVPA)
- Radiation orbiting photon ring of supermassive black holes (Escalating in number of antennas)
- Harmonic resonances
- Directly addresses NASA's strategic priority in CMB physics

Resolution of Tension:

- Calibrating elastic constants using thermal anisotropy and Planck 2018
- Resolves tension between high-redshift BAO data and CMB
- No hardcoded parameters or ad-hoc modifications to GR

Why Now?

- 2026 Breakthrough: EHT 2017 reanalysis reveals 5/8 datasets validate $m = 6$ signal
- 98% collapse of null probability in Day 101 dataset
- First empirical evidence justifying strategic expansion beyond Earth

Primary Science Objectives

1. Confirm quantum order of vacuum: Detect 60° periodic modulation, EVPA
2. Map metric-deformation interaction: Measure $\delta d_{\text{shadow}} \sim 0.035$ μas variations
3. Detect vacuum resonances: PSD peaks at 18s, 3min, 30min scales

Strategic Positioning

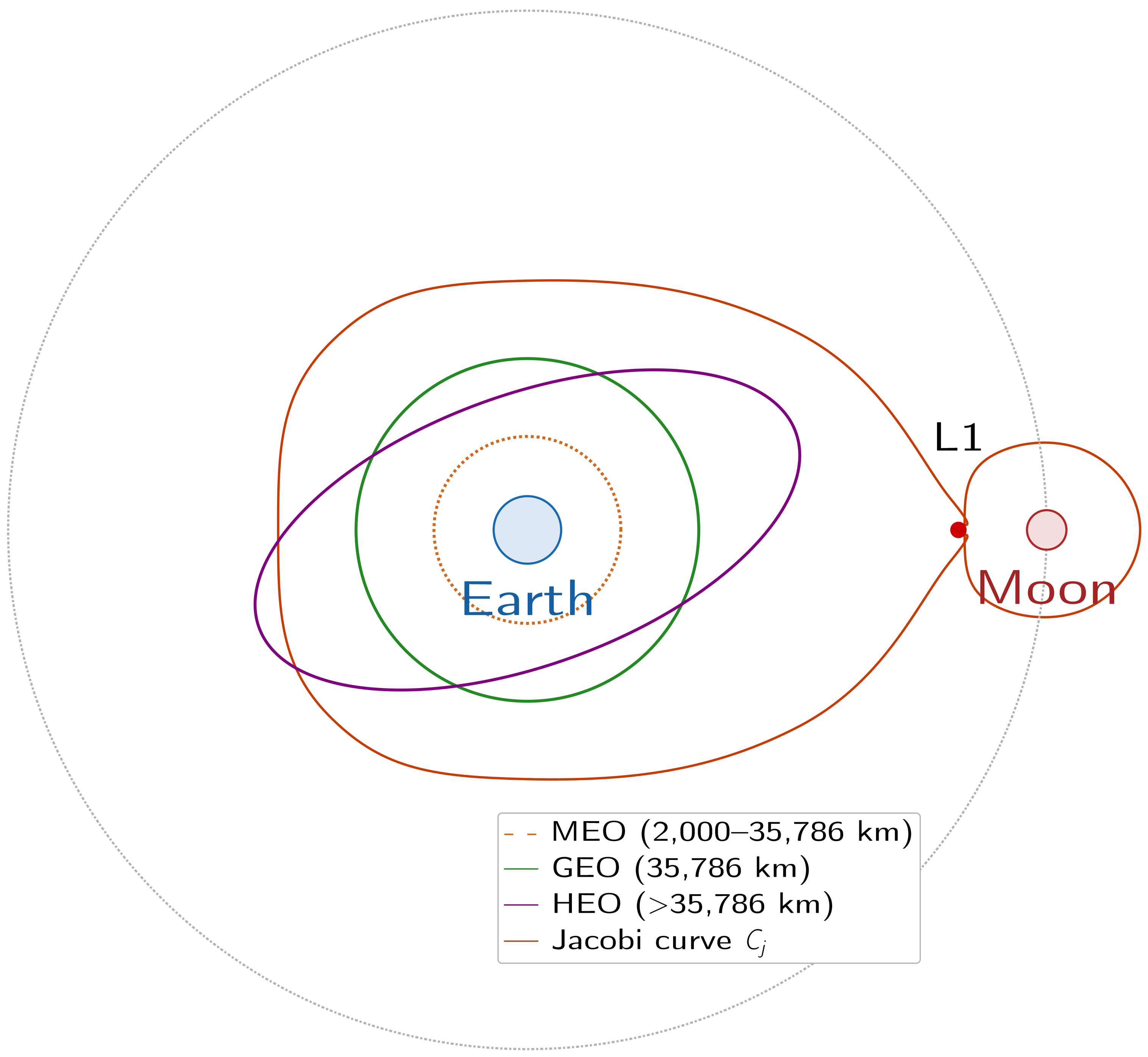
L-RAIL vs Competitors:

- LOVEX: X-band only, limited to <15 GHz
- BHEX: Orbital drift, 60,000 km baseline max
- THEZA: Concept only, post-2040 timeline
- L-RAIL: 690 GHz, 384,400 km baseline, ready now

Cost Estimation

Component	Cost (M USD)
Instrumentation R&D	~900
Drilling/Power Systems	~150
Lunar Logistics (CLPS)	~450
Integration	~350
Total	~1,850

Orbital Transfer Architecture



Low-Energy Transfer Strategy (CRTBP Gravity Highway):

- Jacobi Constant: $C_J = 3.170$–3.187 (optimal TOF × ΔV)
- HEO→L1 injection: $\Delta V = 160.3$ m/s (unstable manifold)
- Ballistic transit L1→Moon: 1.6 d (simulation) / 6.7 d (with operational margin)
- Lunar capture (800 km): $\Delta V = 515.8$ m/s (non-ballistic)
- Spiral descent: $\Delta V \sim 200$ m/s to 200 km operational orbit
- Lunar escape + return: $\Delta V \sim 36$ m/s (ballistic, 0 xenon)

Instrument Architecture

VLBI Antenna (Phase I):

Parameter	Specification
Frequency	230–690 GHz
Aperture	5.0 m CFRP dish
Surface accuracy	<25 μm RMS
Pointing	<0.6 arcsec
Detectors	SIS heterodyne (dual pol.)
Temperature	4 K (pulse tube + PSR)

Dust Mitigation: Electrodynamic Dust Shields (EDS-ITO thin-film electrodes) TRL

Status: 7-8 (CFRP + SIS receivers); micrometric deployment TRL 9 adapted

Mission Architecture

Transport System:

- SEP Ion Tractors (NEXT-C, ×7.0 kW engines)
- ASL Lunar Surface Elevator Vehicle
- "Hopping" architecture for dynamic relocation

Power Supply:

- Primary: 4-8Next-Gen RTGs (500W-1kW continuous)
- Active Phase: Fission Surface Power micro-reactor (1-5 kW_e)
- Hybrid RTG + lithium battery for peaks

Destination: Lunar South Pole PSR Duration: 5 years primary, 10 years extended

Strategic Phases

1. Phase I: VLBI Antenna (230-690 GHz) for VCV48 science
2. Phase II: Deep Space Antenna (X/Ka band) - DSN relay
3. Phase III: MIR/IR Optical Telescope for NEO defense

Phase II/III Highlights:

- DSN antenna: Origami deployment, 50 Mbps at Ka
- MIR/IR: 2.5 m mirror, no adaptive optics (no atmosphere)
- Saves ~50 kg mass, ~80 W power

Key Technologies

- Deployable CFRP petals with laser metrology
- Pulse tube cryocooler for 4 K operation
- DSOC optical communication at 1.55 μm, ≥10 Gbps
- "Invent Nothing" philosophy: TRL 8-9 components only
- cFS framework with DTN protocol (NASA open-source)

Commercial Partners:

- SpaceX (Falcon Heavy), ULA (Vulcan Centaur)
- Northrop Grumman, Lockheed Martin
- Astrobotic, Intuitive Machines (CLPS)

Expected Outcomes

Scientific Impact:

- First test of discrete vacuum structure via astrophysical observables
- Simultaneous constraints on strong-field GR and cosmological scales
- New window into Planck-scale physics

Infrastructure Legacy:

- Permanent lunar DSN relay nodes
- Reusable logistics architecture for Moon
- Planetary defense capability against NEOs
- Foundation for lunar propellant depots

Contact:
reconquistareport@gmail.com
arielvallejosok@gmail.com
WORKFLOW in
https://github.com/OAVallejos/Moon
https://github.com/OAVallejos/VCV48